
Monte Carlo Simulations for Nuclear Reaction Experiments using Silicon Charged Particles Detectors

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ABSTRACT

This study employs the NPTool simulation framework to model nuclear reaction experiments of astrophysical relevance, focusing on the ${}^7\text{Be}(\text{d}, \text{p}){}^8\text{Be}^*$ reaction. Monte Carlo simulations were performed with realistic detector geometries to optimise experimental design and assess data analysis strategies. Simulated results were benchmarked against available measurements to evaluate cross-section behaviour in the low-energy regime. The work demonstrates how NPTool supports precision studies in nuclear astrophysics and outlines the potential of advanced methods, such as *R*-matrix analysis and AI-assisted data processing, to refine future reaction rate determinations within the astrophysically significant Gamow window.

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DECLARATION

The work presented in this report is performed under the guidance of:

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I hereby declare that the report entitled “**Monte Carlo simulations for nuclear reaction experiments using Silicon charged particles detectors**” is based on my original works, inspired by existing literature. I have duly acknowledged all the references that are used partially to prepare this report.

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CHAPTER 1

INTRODUCTION

1.1 Nuclear Reactions

An element itself undergoes a change with change in its nuclear structure, which can be brought about by changing the neutron or proton number or both. To transform a nucleus artificially, a large amount of energy (a few MeV) needs to be provided, at least equal to the binding energy. This energy can be supplied by introducing a nuclear particle in the nucleus. This process of supplying a nuclear particle to the nucleus of a given atom to bring about a transformation in the atom is called a nuclear reaction. The first such nuclear reactions were done in Rutherford's laboratory, using **alpha** particles from a radioactive source.

A typical nuclear reaction is written as $a + X \rightarrow Y + b$ or $X(a,b)Y$ where a is the accelerated projectile, X is the target (usually stationary in the laboratory), and Y and b are the reaction products. Usually, Y is assumed to be a heavy product while b is the lighter particle.

Nuclear reactions can be classified based on the particles involved and based on reaction mechanism. If the incoming and outgoing particles are the same it is a scattering reaction- elastic if the products remain in ground state, else inelastic. If there is a third product, it is a knockout reaction. In direct reactions, only very few nucleons take part in the reaction, with the remaining nucleons of the target serving as passive spectators. The other extreme is the compound reaction, in which the incoming and target nuclei merge briefly for a complete sharing of energy, forming a compound nucleus before the outgoing nucleon is ejected. Between these two states is the resonance reaction- in which the incoming particle forms a "quasibound" state before the outgoing particle is ejected.

1.2 Experimental labs

1. The **Facility for Rare Isotope Beams** in the **United States** is owned by **Michigan State University** and has been in operation since 2022. FRIB supplies a huge variety of heavy ion beams- stable and RIBs.

Some stable target materials at FRIB are:

- **Carbon-12**: for producing neutron rich RIBs.
- **Lead-208**: A heavy target material for producing exotic isotopes of elements like gold and platinum.
- **Uranium-238**: A very heavy target material for studying the production of super-heavy elements.

A beam of really fast stable nuclei is sent barreling through these targets. Nuclei in the beam smash into the target's nuclei, leading to all sorts of nuclear reactions that create unusual and unstable isotopes. A new beam filled with rare and unstable isotopes exits the target. Some of these common RIBs are:

- Neutron rich isotopes like **Calcium-52** and **Oxygen-28**.
- Proton rich isotopes like **Tin-100** and **Phosphorus-26**

In FRIB, Linear Accelerators (**LINAC**) are used most commonly to accelerate ions with a maximum energy of 295 MeV/u. A special type of LINAC called Heavy Ion Linear Accelerator (**HILAC**) is also available.



Fig. 1.1. LINAC at FRIB, US

2. The **Isotope mass Separator On-Line DEvice** facility in Switzerland has been in operation since 1967. It is a unique source of low-energy beams of radioactive nuclides. At its heart of operation is a target and ion source. Over the years, many different advanced

target-ion-source combinations have been developed, which allows the study of radioisotopes from more than 60 different elements.

Some of the targets used at ISOLDE consist of molten metals kept at temperatures from 700°C and up to about 1400°C, having a relatively long release time. Faster release times, in the order of one second or less, can be obtained, if target material in the form of refractory metal powder, metals or carbides is used at temperatures above 2000°C. An expected decrease in the release time due to the "shock-wave" effect of the pulsed proton beam with time constants down to some tenths of a second can be reached. Common target materials:

- **silicon** and **carbon**: Most common
- **Lead**: produce neutron rich isotopes
- **Thorium** and **Uranium** :heavy ion targets, often used in combination

Ion sources used are **surface ion source**, **plasma ion source** and **laser ion source**.



Fig. 1.2. ISOLDE experiment hall

ISOLDE uses common RIBs such as proton rich isotopes of Phosphorus-26 and Tin-100 and neutron rich isotopes of oxygen-28 and calcium-52 with beam energy upto 1.4GeV. Common linear accelerator facilities:

- **HIE-ISOLDE** : a network of High Energy Beam Transfer (HEBT) beamlines- one common beamline, leading to three independent beamlines, all bending the beam by 90° and focusing it using two dipole magnets and a doublet-quadrupole.
- **Offline 2** : mass separator beamline- enclosed in a Faraday cage and laser laboratory and control station, designed for target test studies and study of molecular ion beams.
- **REX-ISOLDE**

3. **Grand Accélérateur National à Ions Lourdes** is a national heavy-ion accelerator facility located in Caen, France. Two major accelerator facilities are available for their experiments-

- **Cyclotron complex**- It consists of five cyclotrons- 2 compact cyclotrons (C01 and C02) with beam energy < 1 MeV/A, 2 separated sector cyclotrons (CSS1 and CSS2) with Beam energies between 13.5 MeV/A and 100 MeV/A and 1 compact cyclotron SPIRAL1 for radioactive beams with Beam energy < 25 MeV/A.
- **SPIRAL2 LINAC**- It is a superconducting linear accelerator that can accelerate lighter nuclei than the cyclotron. It is made of 26 accelerating cavities, enclosed in 19 cryomodules. It also accelerates heavy ions up to nickel, at intensities 10 times greater than those currently available. Energies ranges :
 - 33 MeV for protons
 - 40 MeV for deuterons
 - < 14.5 MeV/A for heavy ions

GANIL provides a variety of beams for experiments:

- Stable beams- occur naturally and relatively abundant. Examples include Carbon-12, Oxygen-16, Calcium-40, Lead-208
- Radioactive beams- unstable isotopes, decay over time. Examples include, Oxygen-15, Phosphorus-32, Iodine-123, Francium-221
- Exotic beams- rare isotopes far from the line of stability. Examples include Beryllium-11, Lithium-11, Carbon-14, Oxygen-28



Fig. 1.3. GANIL, Caen

4. India has a strong foundation in nuclear research. Key institutions like VECC, BARC, and TIFR have made significant contributions to the field.

Variable Energy Cyclotron Centre in Kolkata is renowned for its cyclotrons, its ISOL facility crucial for *rare isotope research*.

Bhaba Atomic Research Centre in Mumbai houses several research reactors like DHRUVA, PURNIMA and APSARA, as well as accelerators for diverse experiments. These reactors are vital for *neutron scattering* and *isotope production*.

Tata Institute of Fundamental Research in Mumbai also has accelerator facilities and specializes in *neutron spectroscopy* and *gamma ray studies*.

So far, we do not have availability of radioactive ion beams in India.



Fig. 1.4. APSARA, BARC, India

CHAPTER 2

TOOLS

2.1 Why Simulation?

In nuclear physics, simulation serves as the first step to designing an experiment. An experiment can only be proposed to carry out once we have an idea of the entire setup and its probable outcomes. The first step to a proposal is understanding the beam availability of the reaction to be carried out- not all experimental facilities have all kinds of beams available. Following this, we try to design a suitable detector setup for the reaction. This is where simulation comes in. Based on the kind of study we plan, we choose our detectors, and configure them at the convenient positions. Different detectors are efficient in different kinds of measurements like position, energy etc, so, we optimize our choice of detectors according to the aim of the experiment. Then we plot the counts and rearrange positions of detectors as required. All this is done through simulation. Fixing the detector angle, beam count, beam energy, angular resolution, etc followed by checking the various parameters through plots- to simulate real experiments is the first step in forming a proposal report.

2.2 Softwares

NPTools, which stands for Nuclear Physics Tool, is an open source and freely distributed data analysis and Monte Carlo simulation package for low-energy nuclear physics experiments. It is a unified framework for preparing and analyzing complex experiments having multiple detectors

and reaction channels, built on CERN-ROOT and GEANT-4. The entire package is written in C++.

It is primarily used for Monte Carlo simulation of reactions, modeling the interaction of particles with matter, including processes like energy loss, scattering, and nuclear reactions. It also models the response of detectors to incident particles, providing a realistic representation of experimental outcomes.

We define input files with extension **.reaction** and **.detector** and run those to create an **executable**. The reaction file contains details of the beam, target, products and the output channels to be simulated. The detector file contains the type of detectors and their arrangements in space. Once we run the executable, the detector setup is simulated in the GEANT4 window. From there, we can run the reaction and that creates the output file. We can open that **output** file through ROOT and plot the various reaction parameters.

CERN-ROOT is a powerful open-source software framework based on C++ primarily used for data analysis in high-energy physics experiments. Developed by CERN, it is used for *simulation and analysis of experimental outputs* and compare with existing literature. All of particle and nuclear physics data analysis tools are based on ROOT.

GEANT4 is also a framework developed for *simulating the passage of particles through matter*. It is majorly used for modeling detector geometry, studying detector response to various beams, MC simulation and modeling physics of particle interactions including electromagnetic, hadronic, and nuclear processes.

CHAPTER 3

SIMULATION

Here, we shall look at simulations for four different reaction setups and their outputs of the following types- Elastic scattering, Inelastic scattering, Transfer and Coulomb breakup reactions.

3.1 $^{208}\text{Pb}(^7\text{Be}, ^7\text{Be})^{208}\text{Pb}$

This is a simple elastic scattering, and is simulated as follows:

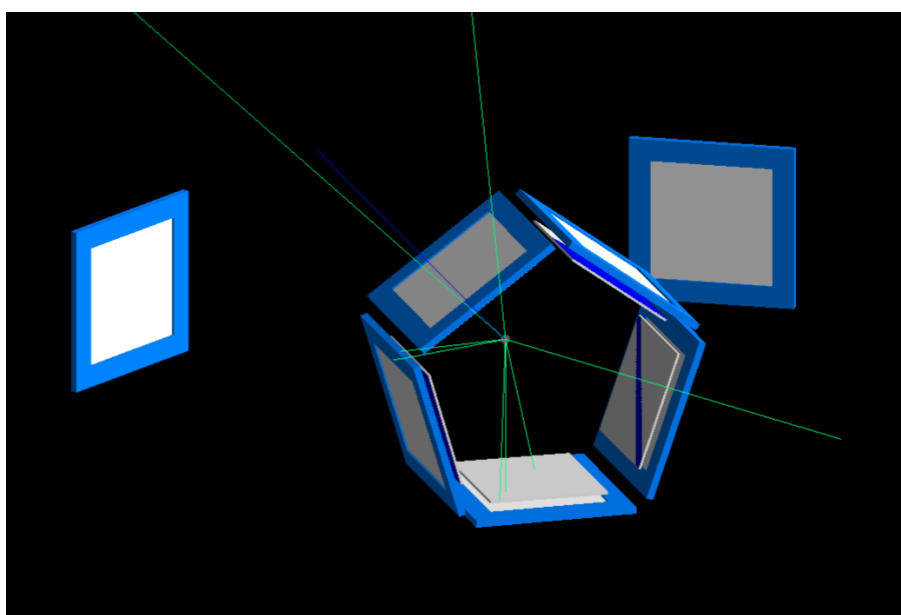


Fig. 3.1. Simulation for the elastic scattering $^7\text{Be} + ^{208}\text{Pb}$ using NPTools

1. A ^7Be beam of energy 35 MeV is impinged on a ^{208}Pb target, with 10000 particle counts.
2. The target has a thickness of $5\mu\text{m}$ and a thickness of 2 mm. We also employ an isotropic cross-section.
3. The detector is composed of 5 double-sided Micron 16×16 silicon strip detectors (DSSD), arranged symmetrically around the target in a regular pentagon, each $60\mu\text{m}$ thick and spaced 59 mm apart.
4. Each of the DSSDs are then layered with single-sided silicon strip detectors (SSSD), for further detection.
5. Two more 32×32 SSSDs are placed in the direction of source to detect rebounds, on the right and left side of the beam.

The angular coverage of this system is around 40° to 80° , given by:

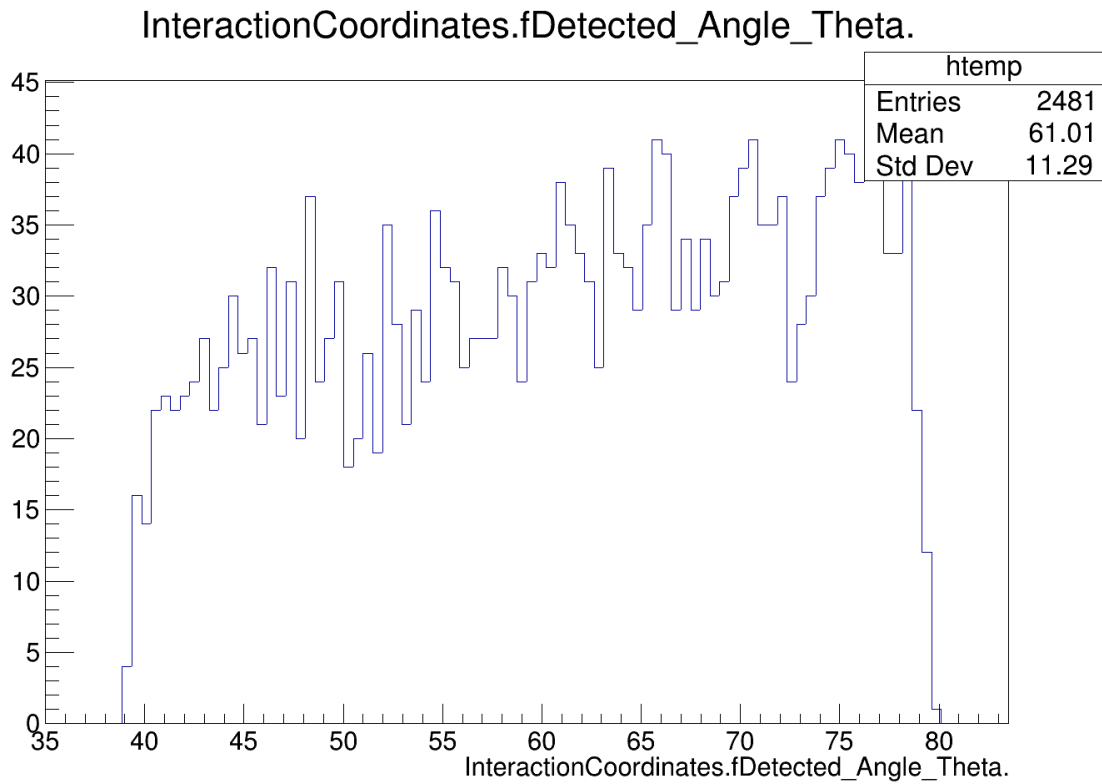


Fig. 3.2. Angular coverage of the system

The reaction kinematics is as follows:

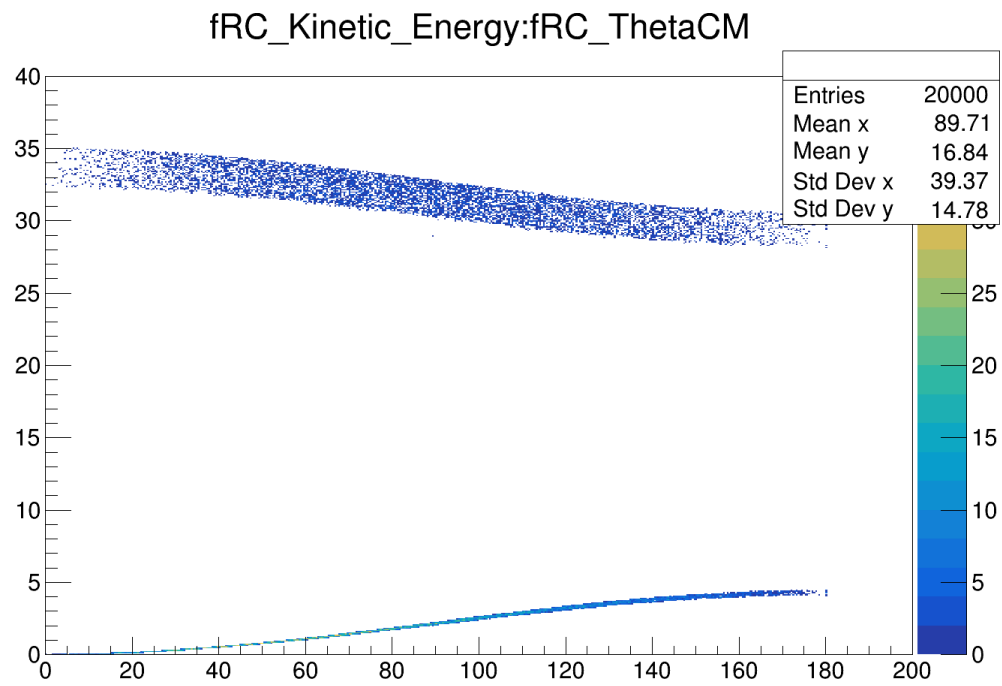


Fig. 3.3. Kinematics of the elastic scattering

The energy deposition of the particles in the detectors with respect to the angular distribution is as follows.

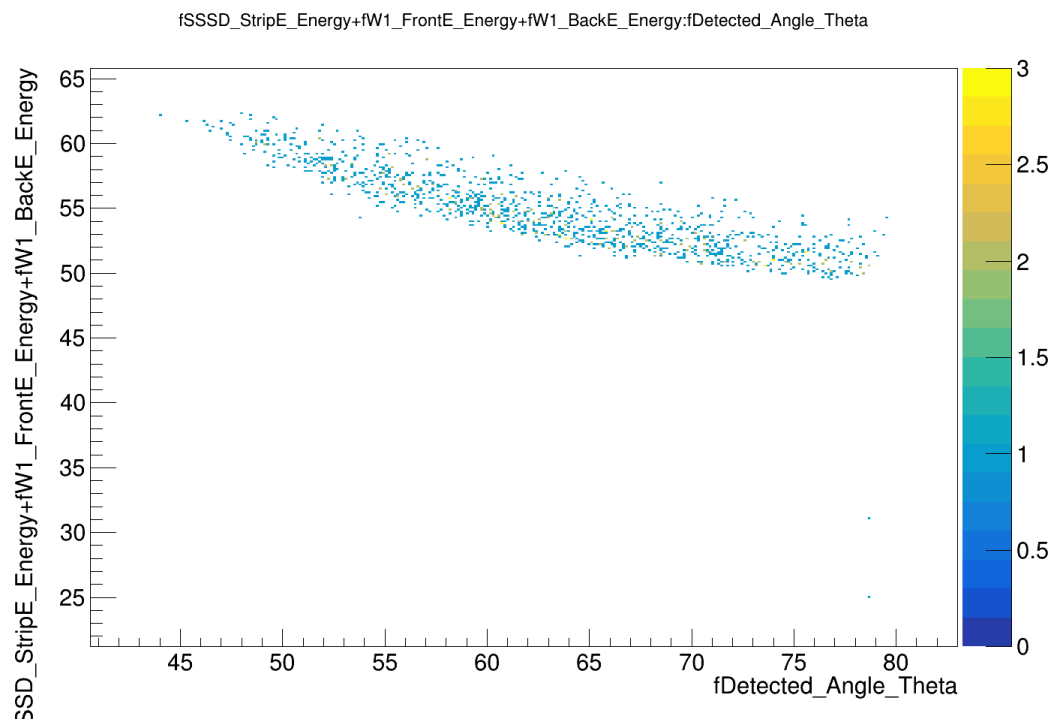


Fig. 3.4. Angular energy deposition of ^7Be

3.2 $^{12}\text{C} (^7\text{Be}, ^7\text{Be}) ^{12}\text{C}^*$

This is an inelastic scattering where the product $^{12}\text{C}^*$ is excited to its first level with excitation energy = 4.44 MeV. The simulation of the reaction is as follows.

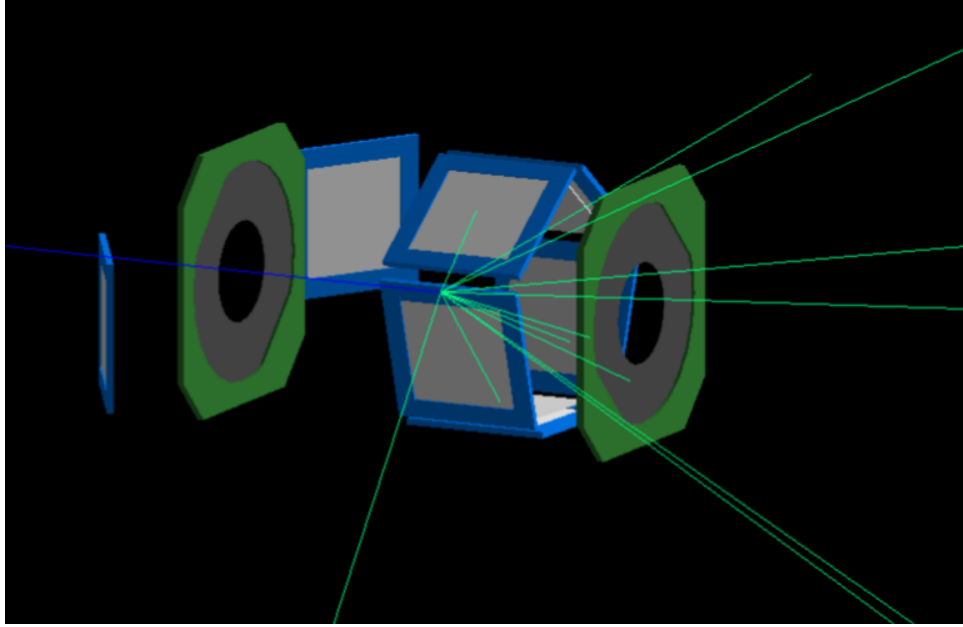


Fig. 3.5. Simulation for the inelastic scattering of $^{12}\text{C} + ^7\text{Be}$ using NPTools

1. A ^7Be beam of energy 35 MeV is impinged on a CD_2 target, with 10000 particle counts.
2. The target has a thickness of $15\mu\text{m}$ and a thickness of 2 mm. We also employ a flat cross section.
3. The heavier product $^{12}\text{C}^*$ is excited to 4.44 MeV.
4. The detector is made of 5 double-sided 16×16 silicon strip detectors (DSSD), placed symmetrically around the target like a regular pentagon, each $60\mu\text{m}$ thick and at a distance of 59 mm.
5. Each of the DSSDs are then layered with single-sided silicon strip detectors (SSSD), for further detection.
6. Two more 32×32 DSSDs called BB7 are placed in the direction of source to detect rebounds, on the right and left side of the beam.
7. Annular S1 detectors with a circular hole are placed in the direction of the beam as well as towards the products. The detector is divided into 32 sectors (called spokes) in the front and 24 rings in the back for position information of the detected particle. The annular disc has an inner radius of 11 mm and outer radius of 35 mm.

The angular coverage of this system is given by:

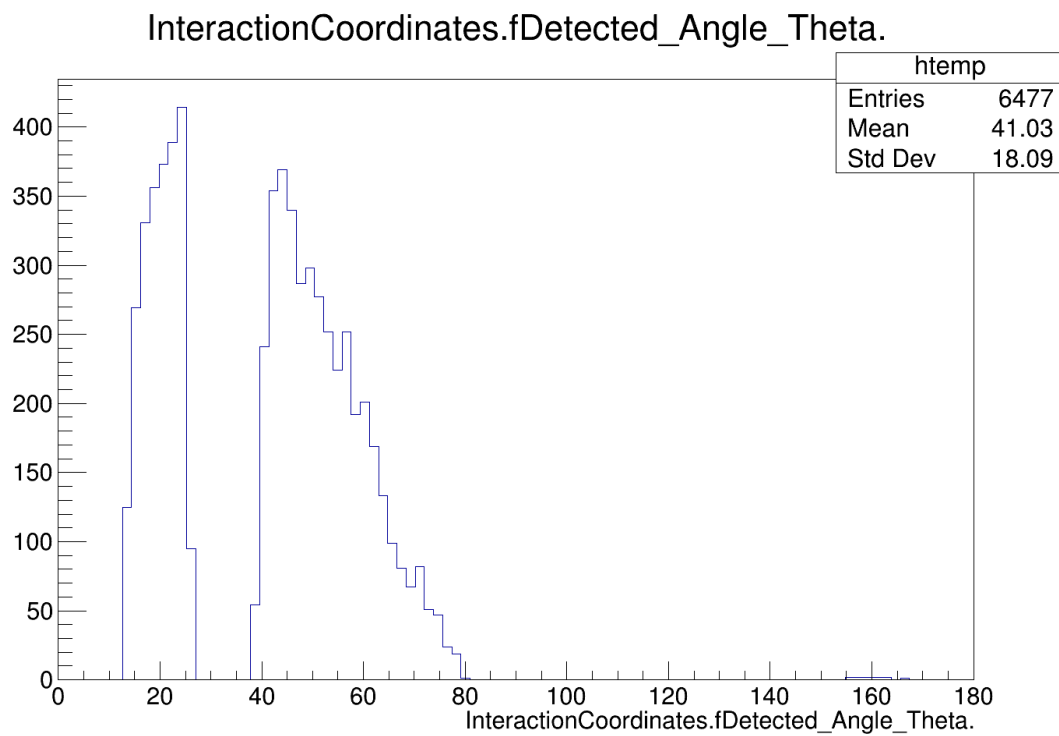


Fig. 3.6. Angular coverage in θ

The energy deposition of the particles with respect to the angular distribution is as follows:

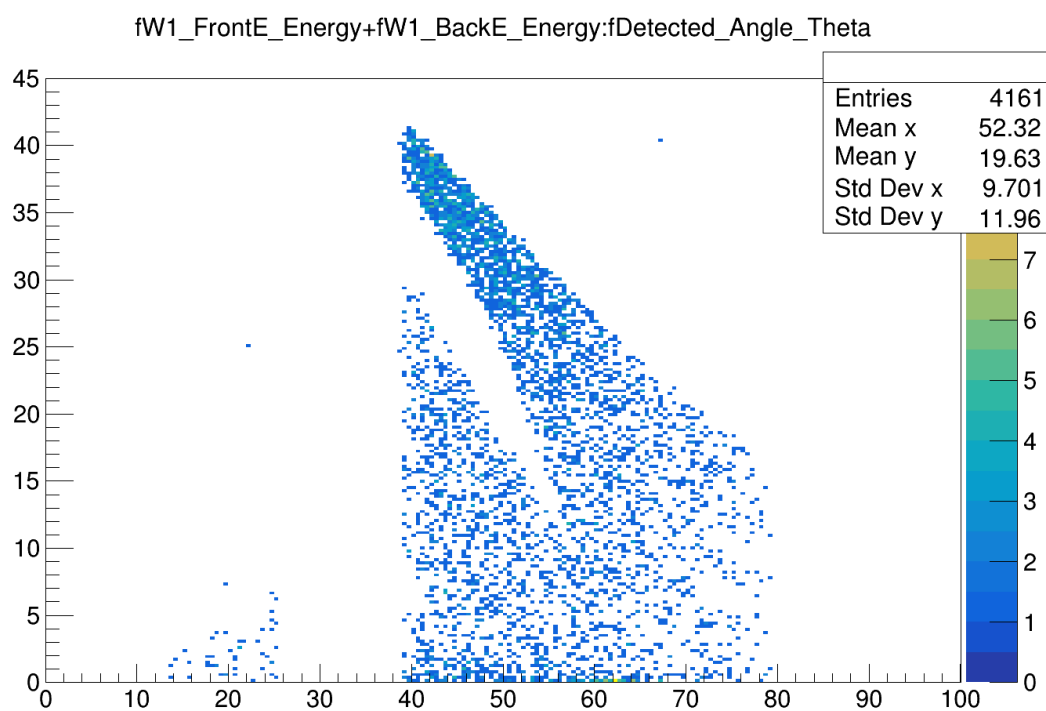


Fig. 3.7. Angular energy deposition in DSSDs

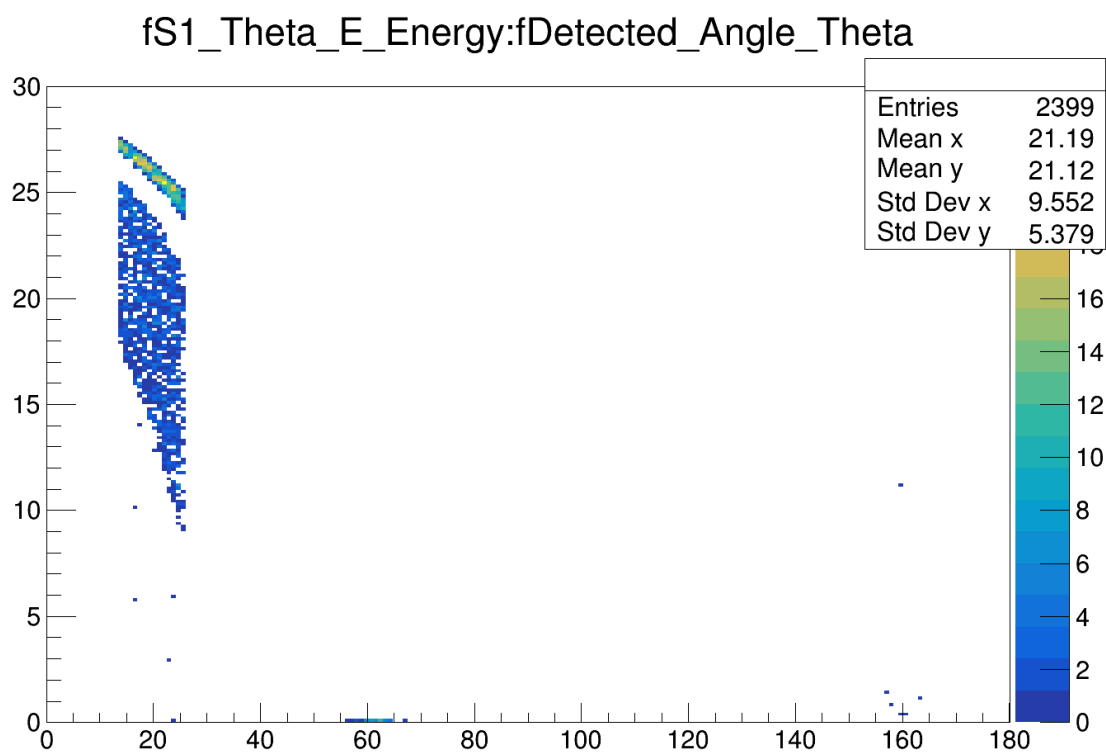


Fig. 3.8. Angular energy deposition in Annular S1

The kinematic of the reaction, is as follows:

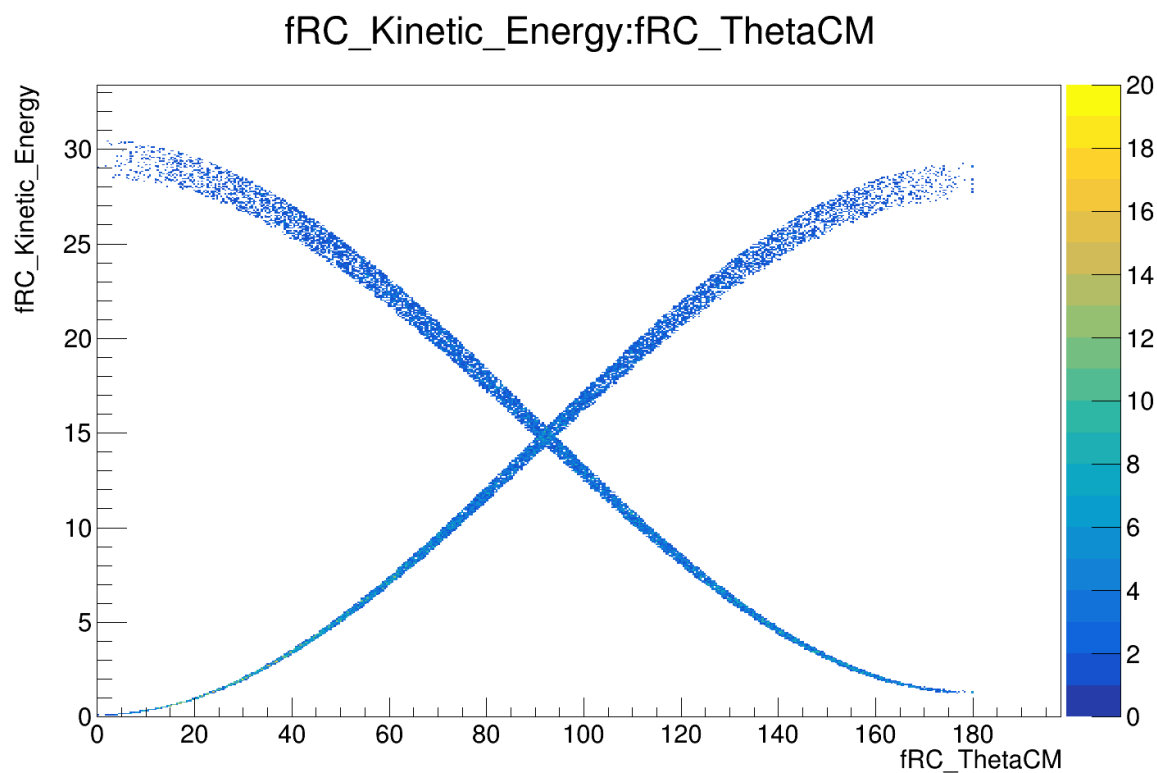


Fig. 3.9. Reaction kinematics of inelastic scattering of $^{12}\text{C} + ^7\text{Be}$

3.3 $^{12}\text{C}(\text{d}, \text{p})^{13}\text{C}$

We create a simulation of experiment of the **Transfer reaction** $^{12}\text{C}(\text{d}, \text{p})^{13}\text{C}$ and study its outputs.

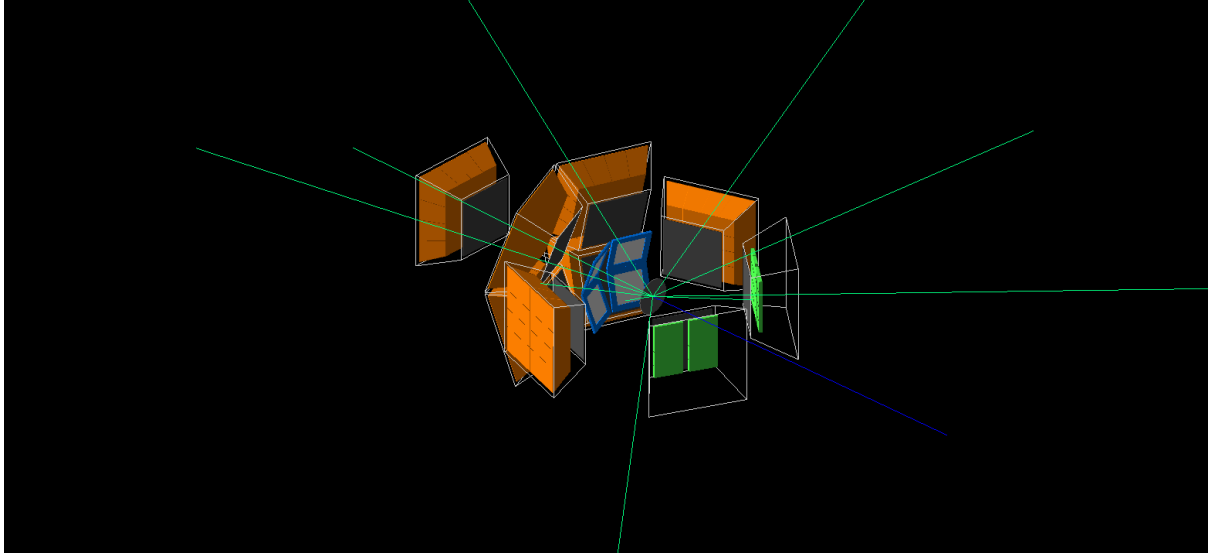


Fig. 3.10. Simulation for $^{12}\text{C}(\text{d}, \text{p})^{13}\text{C}$ using NPTools

1. A d beam at energy 12 MeV is impinging on a ^{12}C target. The products are p and ^{13}C .
2. In this case, we simulate 10000 events.
3. An isotropic cross section is used.
4. The target is made of CD_2 , has a radius of 30 mm and a thickness of $18\ \mu\text{m}$.
5. For detectors, 4 single-sided 16×16 silicon strip detectors (SSSDs) are placed in front of the target at a distance of about 100 mm forming a quadrilateral plate like guard.
6. There are also 7 MUST2 telescopes, with layers of DSSD and CsI only. 4 of them form another layer of detection, behind the SSSDs and 2 more on each side, for detecting particles at wider angles. One is placed further away at the back.
7. Another pair of MUST2 telescopes, having only the DSSD and SiLi is placed close to the source, for detecting rebounds.

This arrangement of the detector gives good angular coverage with maximum detection near smaller angles, as can be studied from the output, plotted using CERN-ROOT.

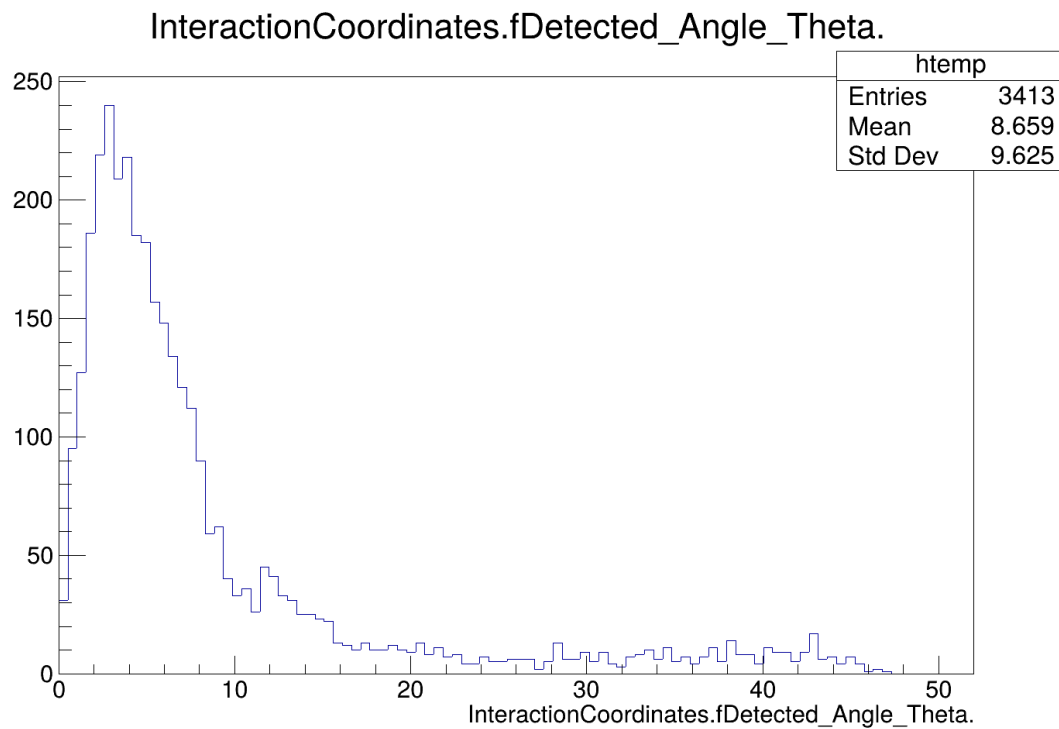


Fig. 3.11. Angular distribution of p using ROOT

Figure 3.12 shows a plot of the energy deposition of the particles with respect to its angular distribution in the SSSD detector, using ROOT. This detector has the maximum detection.

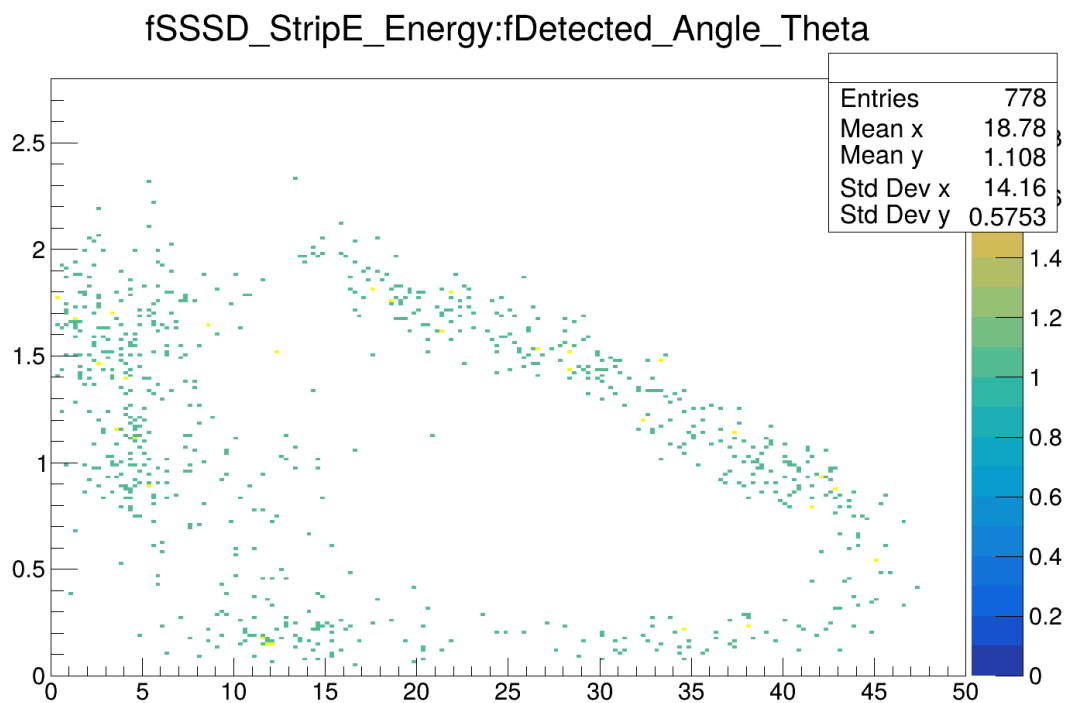
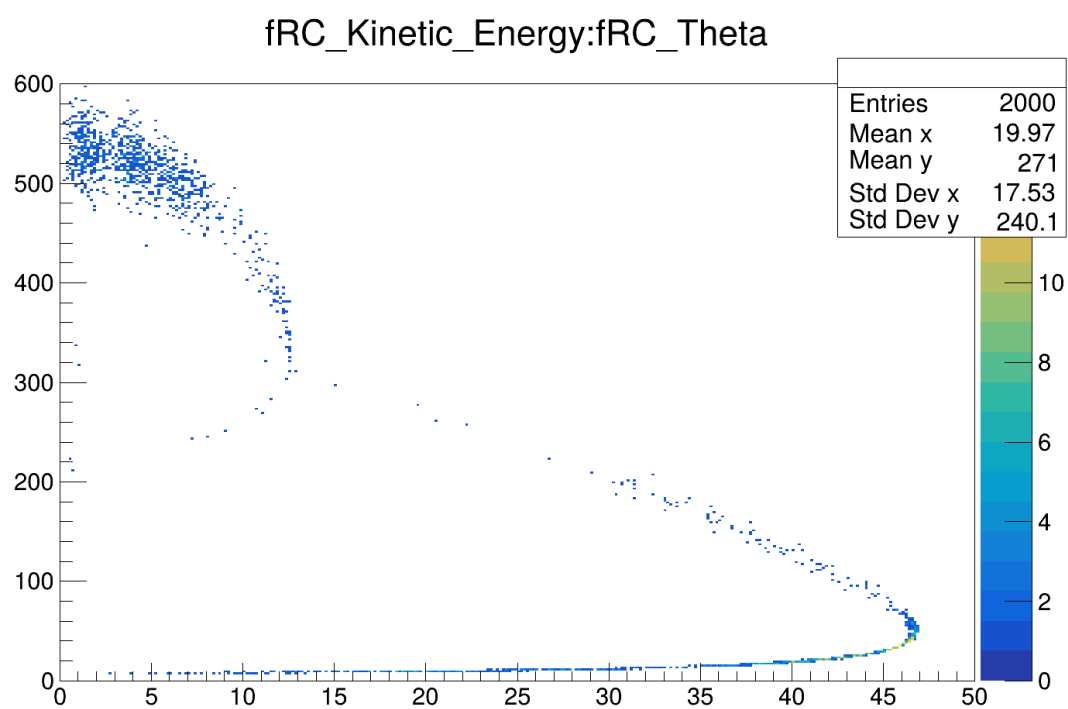


Fig. 3.12. Angular energy deposition of p in SSSD using ROOT

The reaction kinematics gives the energy vs angle plot for a reaction. It is as follows:

**Fig. 3.13.** Reaction Kinematics

3.4 $^{13}\text{N}(p, \gamma) ^{14}\text{O}$

This reaction is a radiative capture reaction relevant in the Hot CNO cycle in nuclear astrophysics. But to study this reaction, we need to perform it backwards. ^{14}O is made to react with ^{208}Pb and we achieve the **coulomb breakup reaction** $^{14}\text{O} + ^{208}\text{Pb} \rightarrow ^{208}\text{Pb} + ^{13}\text{N} + p$. We create a simulation of this reaction as follows:

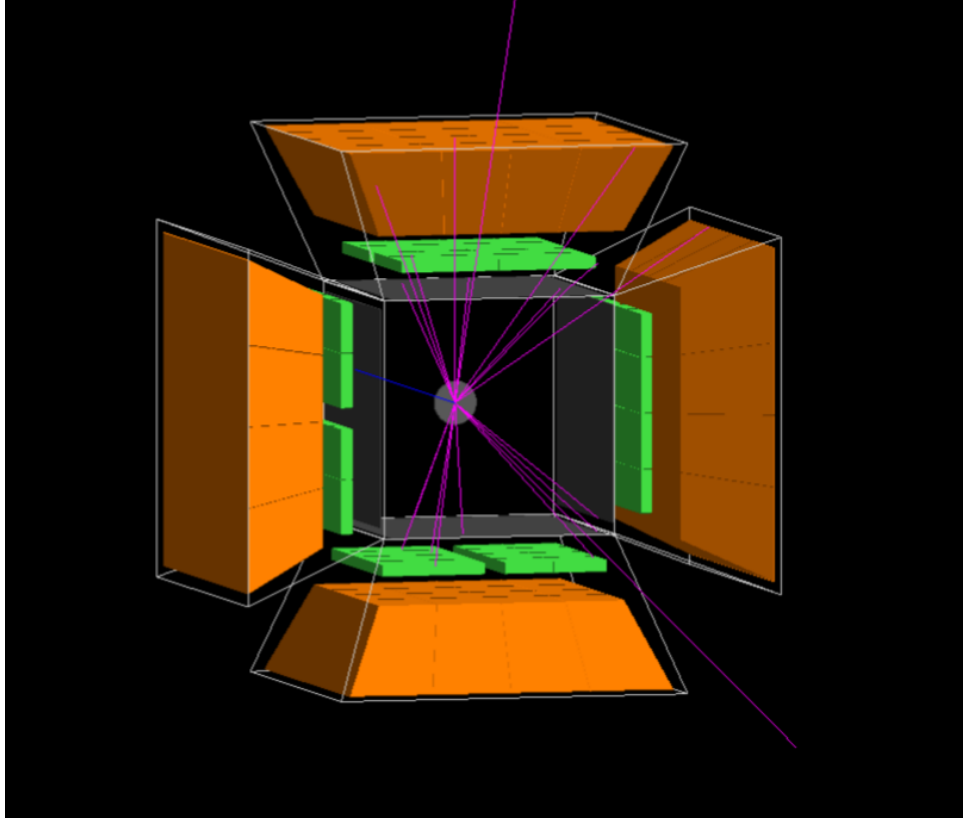


Fig. 3.14. Simulation for the coulomb breakup using NPTools

1. A ^{14}O beam with energy **238 Mev** is impinged on a ^{208}Pb . In this case, the particle count is 10000.
2. The immediate product ^{14}O will split into ^{13}N and p .
3. The target Pb is $5 \mu\text{m}$ thick with a radius of 10 mm.
4. The detector setup is made of 4 pieces of MUST2 telescope arrays, places opposite two each other in the form of a square perimeter around the target at distances of 55 mm. This structure allows the maximum detection of the particles.
5. Each MUST2 array is made of three detection layers:
 - **Double-sided Silicon Strip Detector (DSSD):** $300 \mu\text{m}$ thick, $32 \times 32 \text{ cm}^2$ active area for precise position and energy measurement.

- **SiLi detector** : 3 cm thick for better energy measurement
- **CsI crystal**: 3 cm thick for stopping high-energy particles.

The angular coverage of this setup is roughly from 35° to 110° , as plotted using CERN-ROOT, is as follows:

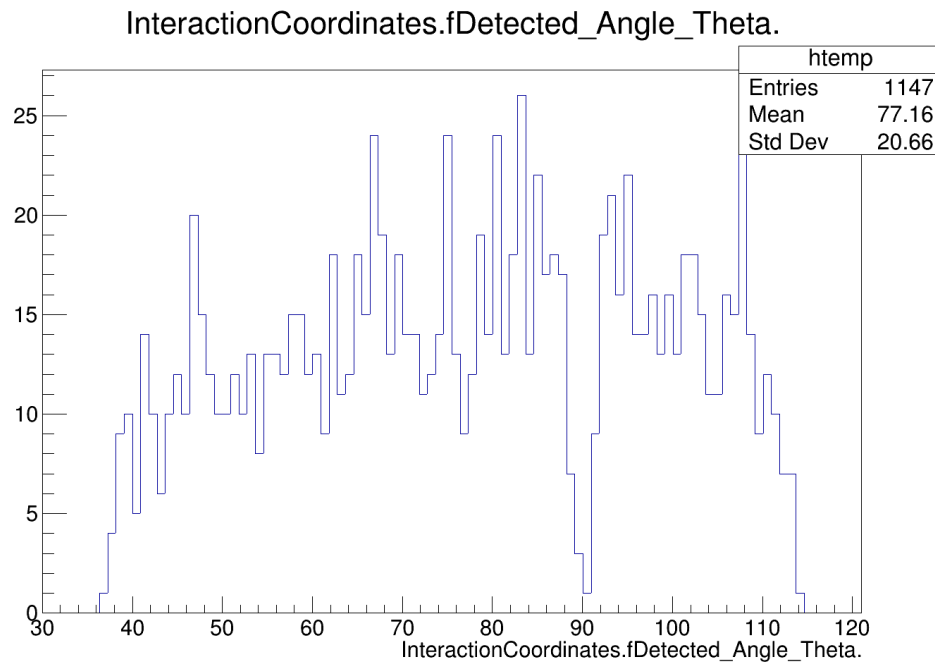


Fig. 3.15. Angular distribution of the particles

The reaction kinematics is given by the information from the Interaction coordinates branch, showing both the particles:

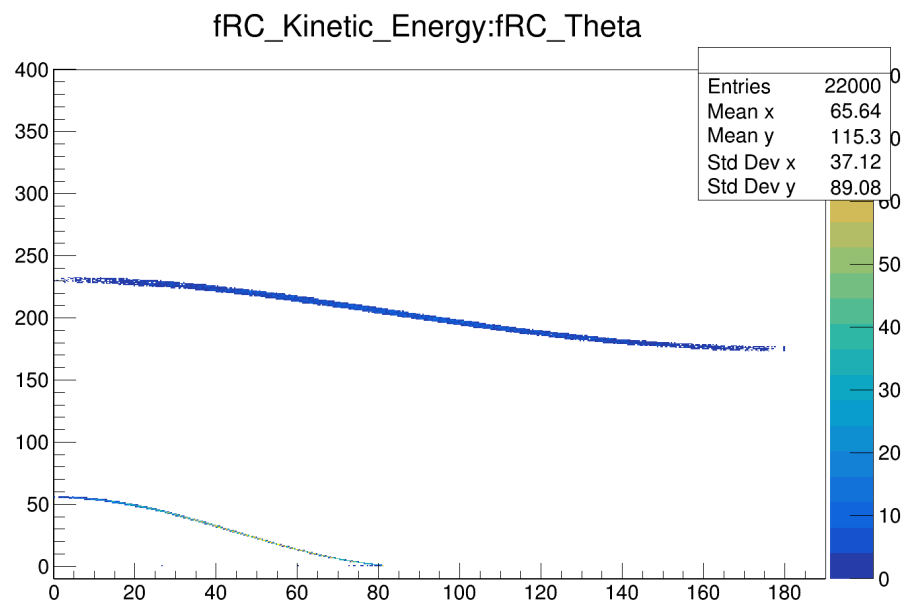


Fig. 3.16. Kinematics of the given reaction

The energy deposition of the particles from the Silicon and SiLi layer with respect to the angular distribution is as follows. Since we had activated both the light and heavy particle channels, we get two distinct lines.

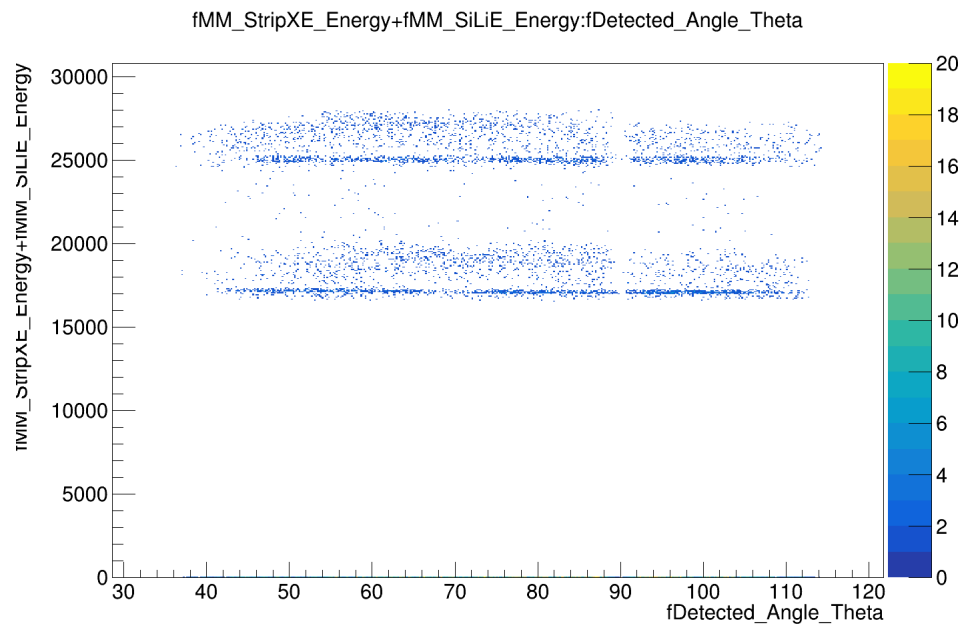


Fig. 3.17. Angular energy deposition of ^{13}N and p

CHAPTER 4

REACTION ${}^7\text{Be} + \text{d} \rightarrow \text{p} + {}^8\text{Be}^*$

4.1 Astrophysical Significance

The ${}^7\text{Be}(\text{d}, \text{p}){}^8\text{Be}^*$ reaction holds critical importance in nuclear astrophysics due to its role in resolving the long-standing cosmological lithium problem. Observations of primordial lithium abundances in metal-poor stars reveal a discrepancy of 3-4 times lower ${}^7\text{Li}$ levels than predicted by Big Bang Nucleosynthesis (BBN) models. Since ${}^7\text{Be}$ (which beta-decays to ${}^7\text{Li}$) is a key intermediate in BBN, its destruction via deuteron-induced reactions could suppress final lithium production. If the ${}^7\text{Be}(\text{d}, \text{p}){}^8\text{Be}^*$ cross-section is significantly enhanced at astrophysical energies (Gamow window, 0.1–1 MeV), this reaction could provide a viable depletion mechanism, reconciling theory with observations. Additionally, in stellar environments, this process competes with ${}^7\text{Be}$ electron capture, influencing the evolution of lithium abundances in low-metallicity stars. Furthermore, since ${}^7\text{Be}$ is linked to the solar neutrino flux via the ${}^7\text{Be}(\text{d}, \gamma){}^8\text{B}$ reaction, understanding its destruction channels is essential for refining solar models. Thus, precise measurements of ${}^7\text{Be}(\text{d}, \text{p}){}^8\text{Be}^*$ cross-sections and S-factors are crucial for both cosmology and stellar astrophysics.

4.2 Possible Attempts at Measurement

Understanding the cross section and S-factor is important in the study of Big Bang Nucleosynthesis, and future attempts have been made, though not with a lot of definite success.

4.2.1 Theoretical Foundations

Barker (1969) and Rolfs (1973) established a theoretical baseline for the (d, p) reaction cross section in light nuclei. Their work provided essential insights into reaction mechanisms and served as a reference framework for subsequent experimental studies. These theoretical models, grounded in direct reaction theory, offered predictions of both the magnitude and energy dependence of the cross section, laying the foundation for later empirical validation.

4.2.2 Advances in Experimental Techniques

The introduction of ISOL (Isotope Separation On-Line) beam technology in the late 1990s marked a significant milestone in experimental nuclear astrophysics. For the first time, these facilities enabled the direct use of radioactive ion beams such as ${}^7\text{Be}$, allowing direct measurement of reaction cross sections that had previously been accessible only through indirect methods.

In 2005, the Cyclotron Research Centre at Louvain-la-Neuve reported preliminary data for the ${}^7\text{Be}(d, p){}^8\text{Be}^*$ reaction at center-of-mass energies $E_{\text{cm}} > 2$ MeV. Although limited in energy coverage and precision, these results represented an important proof of concept for studying this reaction in the laboratory.

4.2.3 Recent Developments

The most notable recent advancement occurred in 2021 at the HIE-ISOLDE facility, CERN, where the first dedicated attempt was made to directly measure the cross section across a range of astrophysically relevant energies. This work leveraged the enhanced beam intensities and improved detector systems available at HIE-ISOLDE, enabling measurements closer to the Gamow window. Such measurements are critical for determining the astrophysical S -factor, which minimizes the exponential suppression from Coulomb barrier effects and provides a clearer picture of the intrinsic nuclear interaction probability.

4.3 Future Perspectives and Advanced Measurements

For the reaction ${}^7\text{Be} + d \rightarrow p + {}^8\text{Be}^*$, recent experimental progress has been significant, yet precise determination of the low-energy cross section and S -factor remains challenging due to the short half-life of ${}^7\text{Be}$ and the low count rates achievable in direct measurements.

Modern computational approaches can address these limitations. Machine Learning (ML) techniques can assist in background suppression, automated peak identification, and rapid parameter optimization for fitting excitation functions. In parallel, R -matrix analysis provides a robust theoretical framework to fit limited datasets and extrapolate the cross section reliably into the astrophysically relevant Gamow window.

By combining ML-driven data processing with *R*-matrix modeling, future measurements of ${}^7\text{Be}(d, p){}^8\text{Be}^*$ can achieve improved precision and reduced uncertainties, leading to better-constrained reaction rates for nucleosynthesis calculations in stellar and Big Bang environments.

CHAPTER 5

CONCLUSION

This project presents a detailed exploration of nuclear reaction simulations using Monte Carlo techniques and silicon detector systems, focusing on elastic, inelastic, transfer, and Coulomb breakup reactions. By employing tools like NPTool, GEANT4, and ROOT, we successfully modeled realistic detector geometries and extracted observables such as angular distributions, energy spectra, and reaction kinematics. NPTool not only served as a powerful simulation framework but also provided a robust environment for event reconstruction and data analysis, making it highly suitable for experimental design and post-run evaluation.

A particular emphasis was placed on the astrophysically significant reaction ${}^7\text{Be}(d, p){}^8\text{Be}^*$. While theoretical baselines were established decades ago, and recent ISOL-based experiments have provided partial data, accurate low-energy cross section and S -factor measurements remain challenging. Modern methods such as Machine Learning for data processing and R -matrix analysis for robust low-energy extrapolation can significantly improve reaction rate estimations within the Gamow window.

Overall, this work demonstrates the combined power of theoretical codes and detector simulations in studying nuclear reactions of astrophysical importance and lays the groundwork for future experiments involving radioactive beams and precision cross-section measurements.

BIBLIOGRAPHY

- [1] Ali, S. M., Gupta, D., Kundalia, K., Saha, S. K., Tengblad, O., Ovejas, J. D., Perea, A., Martel, I., Cederkall, J., Park, J., & Szwec, S. (2022). Resonance excitations in ${}^7\text{Be}(d, p){}^8\text{Be}^*$ to address the cosmological lithium problem. *Physical Review Letters*, 128(25), 252701. <https://doi.org/10.1103/PhysRevLett.128.252701>
- [2] Koning, A., Hilaire, S., & Goriely, S. (2023). TALYS: Modeling of nuclear reactions. *European Physical Journal A*, 59, 131. <https://doi.org/10.1140/epja/s10050-023-01034-3>
- [3] Variable Energy Cyclotron Centre. (2022). *Silicon Detector Laboratory*. Retrieved July 5, 2025, from <https://www.vecc.gov.in/node/629>
- [4] Wikipedia contributors. (2024, May 30). *ISOLDE Solenoidal Spectrometer Experiment*. In *Wikipedia*. Retrieved July 5, 2025, from https://en.wikipedia.org/wiki/ISOLDE_Solenoidal_Spectrometer_experiment
- [5] CERN. (2021). ${}^7\text{Be}(d, p){}^8\text{Be}$ at HIE-ISOLDE. NPTool Meeting 2021. Retrieved July 5, 2025, from <https://indico.cern.ch/event/1068553/contributions/4565568/>
- [6] Wikipedia contributors. (2024, April 28). *TRIUMF*. In *Wikipedia*. Retrieved July 5, 2025, from <https://en.wikipedia.org/wiki/TRIUMF>
- [7] Wikipedia contributors. (2024, June 14). *Laboratori Nazionali di Legnaro*. In *Wikipedia*. Retrieved July 5, 2025, from https://en.wikipedia.org/wiki/Laboratori_Nazionali_di_Legnaro
- [8] Barker, F. C. (1969). Low-energy deuteron stripping in light nuclei. *Australian Journal of Physics*, 22(2), 293–312. <https://doi.org/10.1071/PH690293>
- [9] Rolfs, C. (1973). Deuteron induced reactions on light nuclei. *Nuclear Physics A*, 217(1), 29–70. [https://doi.org/10.1016/0375-9474\(73\)90416-5](https://doi.org/10.1016/0375-9474(73)90416-5)

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- [10] Mukhopadhyay, S., et al. (2023). Advances in R -matrix analysis for nuclear astrophysics. *Journal of Physics G: Nuclear and Particle Physics*, 50(5), 055201. <https://doi.org/10.1088/1361-6471/acbfad>
- [11] Utama, R., Piekarewicz, J., & Prosper, H. B. (2016). Nuclear mass predictions for the crustal composition of neutron stars: A Bayesian neural network approach. *Physical Review C*, 93(1), 014311. <https://doi.org/10.1103/PhysRevC.93.014311>